

Exercises after classes 7-8

Exercise 1. *Measurements of the amount of suspended solids in river water on 14 Monday mornings yield $\bar{x} = 47$, $s = 9.4$. Obtain a 95% confidence interval for the mean amount of suspended solids. State any assumption you make about the population.*

Exercise 2. *A literary critic wants to establish that the mean number of words per sentence, appearing in a newly discovered short story, is different from 9.1 words. A sample of 36 sentences provided the data $\bar{x} = 8.5$ and $s = 1.2$.*

- (a) *Formulate the null and alternative hypotheses. (Define any symbols you use.)*
- (b) *Determine the test statistic.*
- (c) *Give the form of the rejection region.*
- (d) *What is the conclusion to your test? Take $\alpha = .10$.*
- (e) *Calculate a p value.*
- (f) *Based on Part (d), what error could you have possibly made?*

Exercise 3. *A zoologist collected 20 wild lizards in the southwestern United States. The total length (mm) of each was measured.*

179 157 169 146 143 131 159 142 141 130
142 116 130 140 138 137 134 114 90 114

Obtain a 95% confidence interval for the mean length.

Exercise 4. *A genetic model suggests that 80% of the plants grown from a cross between two given strains of seeds will be of the dwarf variety. After breeding 200 of these plants, 136 were observed to be of the dwarf variety.*

- (a) *Does this observation strongly contradict the genetic model?*
- (b) *Construct a 95% confidence interval for the true proportion of dwarf plants obtained from the given cross.*